

COFINITE ZEROS OF HIGH DERIVATIVES

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ABSTRACT. We construct a transcendental entire function for which every nonempty open subset of the complex plane contains a zero of every sufficiently high derivative; equivalently, the zero sets along every strictly increasing sequence of derivative orders have dense union. The construction uses a bounded-coefficient random Fock series and estimates for the probability that a derivative is zero-free in a fixed disk. The resulting function satisfies $|f(z)| \leq \sqrt{2} \exp(|z|^2)$ and provides a counterexample to Theorem 1 of Boas and Reddy as printed; the missing uniformity in their proof is discussed.

1. THE MAIN THEOREM

Throughout, $\mathbb{N} = \{0, 1, \dots\}$. For an entire function f and a nonempty open set $U \subset \mathbb{C}$, put

$$A_U(f) = \{n \in \mathbb{N} : f^{(n)} \text{ has a zero in } U\}.$$

Lemma 1.1. *For an entire function f , the following assertions are equivalent.*

- (1) *For every strictly increasing sequence $(n_k)_{k \geq 0}$ of nonnegative integers, the set*

$$\bigcup_{k \geq 0} \{z \in \mathbb{C} : f^{(n_k)}(z) = 0\}$$

is dense in $\mathbb{C}$.

- (2) *For every nonempty open $U \subset \mathbb{C}$, the set $A_U(f)$ is cofinite in $\mathbb{N}$, meaning that its complement is finite.*

Proof. Assume (2), and fix a strictly increasing sequence (n_k) and a nonempty open set U . There is an $N(U)$ such that every $n \geq N(U)$ belongs to $A_U(f)$. For some k we have $n_k \geq N(U)$, so $f^{(n_k)}$ has a zero in U . The displayed union is therefore dense.

Conversely, if (2) fails, enumerate the infinite set $\mathbb{N} \setminus A_U(f)$ increasingly as (n_k) for a suitable nonempty open set U . The zero set of each $f^{(n_k)}$ misses U , so their union is not dense. $\square$

An entire function is called transcendental if it is not a polynomial.

Theorem 1.2. *There exists a transcendental entire function f such that, for every nonempty open set $U \subset \mathbb{C}$, there exists $N(U)$ with*

$$n \geq N(U) \implies f^{(n)} \text{ has a zero in } U.$$

The same function can be chosen to satisfy

$$(1) \quad |f(z)| \leq \sqrt{2} \exp(|z|^2) \quad (z \in \mathbb{C}).$$

Consequently, the zero sets of the derivatives indexed by every strictly increasing sequence have dense union.

This answers the transcendental version of Erdős Problem #906 [1]; excluding polynomials removes the trivial case. By Lemma 1.1, this is equivalent to the subsequence formulation. We obtain f from a bounded random Fock series. For each fixed disk, the probability that $f^{(n)}$ is zero-free there is summable in n , and a countable basis completes the construction.

2. THE RANDOM FOCK SERIES

Let $\xi_0, \xi_1, \dots$ be independent random variables, each uniformly distributed on the closed unit disk $\overline{\mathbb{D}}$, with normalized planar area as their common law; see [5] for background on bounded random analytic series. Define

$$(2) \quad f(z) = \sum_{k=0}^{\infty} \xi_k \frac{z^k}{\sqrt{k!}}.$$

Since $|\xi_k| \leq 1$, the series converges normally on compact subsets of $\mathbb{C}$ for every sample, and hence defines an entire function that may be differentiated termwise. Writing $F_n = f^{(n)}$ gives

$$(3) \quad F_n(z) = \sum_{m=0}^{\infty} \xi_{n+m} \frac{\sqrt{(n+m)!}}{m!} z^m.$$

The coefficient law has no atoms. Almost surely every ξ_k is nonzero, so f is not a polynomial.

Cauchy–Schwarz with weights $2^{k/2}$ gives

$$\begin{aligned} |f(z)| &\leq \sum_{k \geq 0} \frac{|z|^k}{\sqrt{k!}} \\ &\leq \left(\sum_{k \geq 0} \frac{2^k |z|^{2k}}{k!} \right)^{1/2} \left(\sum_{k \geq 0} 2^{-k} \right)^{1/2} = \sqrt{2} \exp(|z|^2). \end{aligned}$$

Thus (1) holds deterministically. In particular, the resulting function has order at most 2 and finite type.

For $r \geq 0$ set

$$a_{n,m}(r) = \frac{\sqrt{(n+m)!}}{m!} r^m, \quad S_n(r) = \sum_{m=0}^{\infty} a_{n,m}(r),$$

and define the potential and loss by

$$(4) \quad p_n(z) = \frac{1}{2} \log(n!) + |z|\sqrt{n}, \quad L_n = 2 + \frac{1}{2} \log n.$$

For logarithms of moduli we use the standard extended-real convention $\log 0 = -\infty$.

Lemma 2.1 (Saddle estimates). *Let $n \geq 1$.*

(1) *If $r \geq 0$, $1 \leq r\sqrt{n}$, and $r \leq \sqrt{n}$, then for $m = \lfloor r\sqrt{n} \rfloor$,*

$$\log a_{n,m}(r) \geq \frac{1}{2} \log(n!) + r\sqrt{n} - 2 - \frac{1}{2} \log n.$$

(2) *For every $r \geq 0$,*

$$\log S_n(r) \leq \frac{1}{2} \log(n!) + r\sqrt{n} + \frac{r+r^2}{2}.$$

Consequently, if $|\xi_k| \leq 1$ and $|z| \leq R$, then

$$(5) \quad \log |F_n(z)| - p_n(z) \leq \frac{R+R^2}{2},$$

Proof. Factor $a_{n,m}(r) = \sqrt{n!} b_{n,m}(r)$, where

$$b_{n,m}(r) = \frac{\sqrt{(n+1)(n+2) \cdots (n+m)}}{m!} r^m.$$

Put $x = r\sqrt{n}$. Then $b_{n,m}(r) \geq x^m/m!$. For $m = \lfloor x \rfloor$, the assumptions give $1 \leq m \leq x \leq n$. The elementary Stirling bound

$$\log(m!) \leq m \log m - m + \frac{1}{2} \log m + 1$$

implies

$$\log \frac{x^m}{m!} \geq m \log \frac{x}{m} + m - \frac{1}{2} \log m - 1 \geq x - 2 - \frac{1}{2} \log n,$$

where the last inequality uses $m \log(x/m) \geq 0$, $m \geq x - 1$, and $\log m \leq \log n$.

For (2), write

$$b_{n,m}(r) = \sqrt{\frac{(n+m)}{n}} \frac{r^m}{m!}.$$

If $r > 0$, put $t = r/(r + \sqrt{n}) \in (0, 1)$. Cauchy–Schwarz and the negative-binomial and exponential series give

$$\begin{aligned} \sum_{m \geq 0} b_{n,m}(r) &\leq \left(\sum_{m \geq 0} \binom{n+m}{n} t^m \right)^{1/2} \left(\sum_{m \geq 0} \frac{(r^2/t)^m}{m!} \right)^{1/2} \\ &= (1-t)^{-(n+1)/2} \exp\left(\frac{r^2}{2t}\right). \end{aligned}$$

Since $\log(1+u) \leq u$ for $u \geq 0$ and $\sqrt{n} \geq 1$,

$$\frac{n+1}{2} \log \left(1 + \frac{r}{\sqrt{n}} \right) + \frac{r(r+\sqrt{n})}{2} \leq r\sqrt{n} + \frac{r+r^2}{2}.$$

For $r = 0$ the inequality is immediate. Equation (3) and the triangle inequality give $|F_n(z)| \leq S_n(|z|)$, and hence (5). $\square$

For $a \neq 0$, $w \in \mathbb{C}$, and $\varepsilon \geq 0$, normalized area on $\mathbb{D}$ satisfies

$$(6) \quad \mathbb{P}\{|a\xi + w| < \varepsilon\} \leq \left(\frac{\varepsilon}{|a|} \right)^2.$$

Indeed, the preimage in $\mathbb{D}$ is contained in a disk of radius $\varepsilon/|a|$, and normalization cancels the factor π .

Lemma 2.2. *Suppose $1 \leq |z|\sqrt{n}$ and $|z| \leq \sqrt{n}$. For $s \geq 0$,*

$$(7) \quad \mathbb{P}\{p_n(z) - \log |F_n(z)| > s + L_n\} \leq e^{-2s}.$$

Proof. Let $m = \lfloor |z|\sqrt{n} \rfloor$ and condition on all coefficients except ξ_{n+m} in (3). Then

$$F_n(z) = a\xi_{n+m} + w, \quad |a| = a_{n,m}(|z|).$$

Lemma 2.1(1) gives $\log |a| \geq p_n(z) - L_n$. The event in (7) is contained in $\{|a\xi_{n+m} + w| < |a|e^{-s}\}$. Apply (6) conditionally and integrate over the remaining coefficients. $\square$

Lemma 2.3. *Let $Y(\omega, \theta) \geq 0$ be jointly measurable for $0 \leq \theta \leq 2\pi$, and suppose*

$$\mathbb{P}\{Y(\cdot, \theta) > s\} \leq e^{-s/2} \quad (s \geq 0)$$

uniformly in θ . If $\bar{Y} = (2\pi)^{-1} \int_0^{2\pi} Y(\cdot, \theta) d\theta$, then for $t \geq 8$,

$$\mathbb{P}\{\bar{Y} \geq t\} \leq 2e^{-t/8}.$$

Proof. The tail hypothesis gives integrability. Put $E = \{\bar{Y} \geq t\}$ and $q = \mathbb{P}(E)$. Tonelli's theorem gives

$$tq \leq \frac{1}{2\pi} \int_0^{2\pi} \mathbb{E}(Y(\cdot, \theta) \mathbf{1}_E) d\theta.$$

Applying the layer-cake formula to E , we obtain, uniformly in θ ,

$$\mathbb{E}(Y(\cdot, \theta) \mathbf{1}_E) \leq \int_0^\infty \min\{q, e^{-s/2}\} ds = 2q \log \frac{e}{q}.$$

If $q > 0$, then $t \leq 2 \log(e/q)$, and hence $q \leq e^{1-t/2} \leq 2e^{-t/8}$ for $t \geq 8$. There is nothing to prove when $q = 0$. $\square$

3. ZERO-FREE DISKS

For $c \in \mathbb{C}$ and $r > 0$, define

$$\gamma(c, r) = \frac{1}{2\pi} \int_0^{2\pi} |c + re^{i\theta}| d\theta - |c|.$$

Strict convexity of the Euclidean norm gives

$$(8) \quad \gamma(c, r) > 0.$$

Indeed, the triangle inequality gives $|c + re^{i\theta}| + |c - re^{i\theta}| \geq 2|c|$ for every θ . This inequality is strict for some θ : it is immediate when $c = 0$, and when $c \neq 0$ one may take $re^{i\theta}$ perpendicular to c . Continuity makes the inequality strict on an interval. Integration, followed by the change $\theta \mapsto \theta + \pi$, proves (8).

Fix $c \neq 0$ and $0 < R < |c|$, and put $r = R/2$. On the circle $|z - c| = r$,

$$0 < \delta := |c| - r \leq |z| \leq |c| + r =: M.$$

For all sufficiently large n , both $1 \leq \delta\sqrt{n}$ and $M \leq \sqrt{n}$, so the hypotheses of Lemma 2.2 hold uniformly on this circle. To keep the circle deficit real-valued, put

$$\ell(w) = \begin{cases} \log |w|, & w \neq 0, \\ 0, & w = 0. \end{cases}$$

The function ℓ is Borel measurable. Define

$$D_{n,z} = \max\{p_n(z) - \ell(F_n(z)) - L_n, 0\}, \quad \bar{D}_n = \frac{1}{2\pi} \int_0^{2\pi} D_{n,c+re^{i\theta}} d\theta.$$

Since $\ell(w) \geq \log |w|$ in the extended-real order, for $s \geq 0$,

$$\{D_{n,z} > s\} \subset \{p_n(z) - \log |F_n(z)| > s + L_n\}.$$

Lemma 2.2 therefore gives the section tail required by Lemma 2.3. The map $(\omega, \theta) \mapsto D_{n,c+re^{i\theta}}(\omega)$ is jointly measurable, since the random derivative field is jointly measurable and ℓ is Borel measurable. The uniform pointwise tail gives integrability. Hence Lemma 2.3 applies to $\overline{D}_n$.

Let $H_n(c, R)$ be the set of samples for which F_n has no zero in $B(c, R)$. This event is measurable. Indeed, it is the intersection, over rational $0 < \rho < R$, of the events

$$\left\{ \min_{|z-c| \leq \rho} |F_n(z)| > 0 \right\}.$$

For fixed ρ , continuity in z allows the minimum to be computed as the infimum over the countable dense set $(\mathbb{Q} + i\mathbb{Q}) \cap \overline{B(c, \rho)}$; the corresponding point evaluations are measurable. On this set F_n is zero-free on $\overline{B(c, r)}$, and Jensen's formula [4, Chapter III] gives

$$(9) \quad \frac{1}{2\pi} \int_0^{2\pi} \log |F_n(c + re^{i\theta})| d\theta = \log |F_n(c)|.$$

Lemma 2.1(2) gives

$$\log |F_n(c)| \leq p_n(c) + C(c), \quad C(c) = \frac{|c| + |c|^2}{2}.$$

On $H_n(c, R)$, the function F_n has no zeros on $\overline{B(c, r)}$, so $\ell(F_n(z)) = \log |F_n(z)|$ there. Thus $D_{n,z} \geq p_n(z) - \log |F_n(z)| - L_n$, and equations (8) and (9) imply

$$\begin{aligned} \overline{D}_n &\geq \frac{1}{2\pi} \int_0^{2\pi} p_n(c + re^{i\theta}) d\theta - p_n(c) - C(c) - L_n \\ &= \gamma(c, r)\sqrt{n} - C(c) - 2 - \frac{1}{2} \log n =: T_n. \end{aligned}$$

The measurable event $G_n = \{\overline{D}_n \geq T_n\}$ contains $H_n(c, R)$. Moreover, $T_n \rightarrow \infty$ and $T_n \geq \frac{1}{2}\gamma(c, r)\sqrt{n}$ for all sufficiently large n . Lemma 2.3 gives

$$(10) \quad \mathbb{P}(H_n(c, R)) \leq \mathbb{P}(G_n) \leq 2e^{-T_n/8} \leq 2 \exp\left(-\frac{\gamma(c, r)}{16}\sqrt{n}\right)$$

for all sufficiently large n . The series on the right converges, and hence

$$(11) \quad \sum_{n=0}^{\infty} \mathbb{P}(H_n(c, R)) < \infty.$$

By the first Borel–Cantelli lemma, almost surely only finitely many F_n are zero-free in $B(c, R)$.

Let $\mathcal{B}$ be the countable family of disks $B(c, R)$ with $c \in \mathbb{Q} + i\mathbb{Q}$, $R \in \mathbb{Q}_{>0}$, and $R < |c|$. Every nonempty open set contains an element of $\mathcal{B}$: choose a nonzero point in the open set, then a nearby rational center and a sufficiently small positive rational radius.

For each $B \in \mathcal{B}$, almost surely only finitely many F_n are zero-free in B . Since $\mathcal{B}$ is countable, this holds simultaneously for all $B \in \mathcal{B}$ on an event of probability one. Intersecting with the probability-one event on which every ξ_k is nonzero leaves a probability-one event.

Choose a sample in this intersection and denote its entire function by f . It is nonzero and nonpolynomial. If $U \subset \mathbb{C}$ is nonempty and open, choose $B \in \mathcal{B}$ with $B \subset U$. Every sufficiently high derivative of f has a zero in B , and hence in U .

4. THE THEOREM OF BOAS AND REDDY

In their Bulletin announcement, Boas and Reddy state that if an entire function has order at most 2 and finite type, then there is, somewhere in the plane, an arbitrarily large disk in which $f^{(k)}$ is zero-free for infinitely many values of k [2, Theorem 1]. Their discussion on the following page emphasizes that the disk is fixed for those infinitely many derivatives. The expanded article [3] repeats the statement and gives a proof.

Theorem 1.2 provides a counterexample to the statement of Theorem 1 in [2] as printed. For any fixed nonempty open disk D , there is an $N(D)$ such that $f^{(n)}$ has a zero in D whenever $n \geq N(D)$. Only finitely many derivatives are therefore zero-free on D , while (1) places f in the asserted growth class.

Theorem 4.1. *The statement of Theorem 1 in [2], as printed, is false.*

The proof does not establish the required uniformity at the following step [3, p. 469]. In their notation, after fixing a radius c_1 , the negation of the theorem gives an eventual zero-hitting assertion for every fixed disk of radius c_1 . For a fixed R , finitely many such disks are packed into $|z| < R$, so one may take the maximum of their finitely many exceptional thresholds. This maximum may depend on R . The proof then chooses $R = (kd)^{1/2}$ while k ranges through an infinite sequence supplied by a zero-count estimate. No argument shows that a selected k is beyond the threshold belonging to this expanding radius and its moving family of disks. A pointwise eventual assertion for each fixed disk is thus used as if it were uniform over an expanding family. The required uniformity is not established.

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